CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Master of Business Administration (MBA)
Semester – I, University Examination 2009 – 2010
MB 701 Quantitative Analysis for Management - I

Date: 05/01/2010, Tuesday **Time: 01.00 pm – 04.00 pm** **Max. Marks: 70**

Instructions:
1. **This paper contains six questions, of which last one is a case analysis.**
2. **All questions are compulsory**
3. **Each of the first five questions will carry 10 marks and the last case analysis will carry 20 marks.**
4. **Statistical table would be provided on request and students are allowed to carry only scientific calculator. However, exchange of statistical table/scientific calculator is not allowed.**

Q1. (a) A behavioural scientist is conducting a survey to determine if the financial benefits in terms of salary influence the level of satisfaction of employees or whether there are other factors such as work environment which are more important than salary in influencing employee satisfaction. A random sample of 300 employees is given a test to determine their level of satisfaction. Their salary levels are also recorded and presented in tabulated format as below.

	Annual Salary (in Rs Lakhs)			**Total**
Level of Satisfaction	**Upto 5L**	**5L – 10L**	**> 10 L**	
High	10	10	10	30
Medium	50	45	15	110
Low	40	15	5	60
Total	**100**	**70**	**30**	**200**

At 5% level of significance, determine whether the level of employee satisfaction is influenced by salary level. **(5M)**

(b) As a head of the department of a consumer's research organization, you have the responsibility for testing and comparing lifetimes of four brands of electric bulbs. Suppose you test the life-time of three electric bulbs of each of the four brands. The data are shown below, each entry representing the lifetime of an electric bulb, measured in hundreds of hours.

BRAND			
A	B	C	D
20	25	24	23
19	23	20	20
21	21	22	20

Can we infer that the mean lifetimes of the four brands of electric bulbs are equal? **(5M)**

Q2. (a) In an automobile, the shaft (it converts traslatory motion to rotatory motion) can fail either due to failure of bearing or failure of crankshaft mechanisms. The probability of failure of bearing and crankshaft are 0.2 and 0.3 respectively and the probabilities of failure of shaft due to failure of bearing and due to failure of crankshaft are 0.5 and 0.8 respectively. If the shaft has also failed in an automobile, what are the probabilities that it failed due to either by failure of bearing or by crankshaft? **(5M)**

(b) The sectoral growth rate in bank credit as on Oct 2006 has been as follows. It may be noted that the growth in different sectors has varied from 20.3% to 88%. Can we measure the variation in these growths (%)? If so, suggest a suitable measure and calculate it.

Sector	Growth (%)
Agriculture	30.8
Loans to Industry	24.8
Small Scale Industries	20.3
Retail loans	34.3
Housing	32.3
Credit cards	51.1
Education	53.7
Transport operators	88.0
Loans to professionals	48.6
Loans to trade	39.2
Real Estate loans	83.9
NBFCS	28.8

Source: Economic Times, dt 20th Feb 2007 **(5M)**

Q3. (a) A sales manager of a firm believes that 50% of the firm's orders come from the first time customers. A simple random sample of 100 orders will be used to estimate the proportion of first-time customers. Assume that the sales manager is correct and proportion is 0.30

(i) Justify sampling distribution of proportion for this case.

(ii) What is probability that the sample proportions will be between 0.20 and 0.40? **(5M)**

(b) The personnel department of an organization would like to estimate the family dental expense of its employees to determine the feasibility of providing a dental insurance plan. A sample of 10 employees reveals the following family dental expenses (in thousand Rs) in the previous year.

11, 37, 25, 62, 51, 21, 18, 43, 32, 20

Set a 99% confidence interval of the average family dental expenses for the employees of the organization. **(5M)**

Q4. (a) A toy manufacturer is considering a project for manufacturing a dancing doll with three different movement designs. The doll will be sold at an average price of Rs 10. The first movement design using gears and levels will provide the lowest tooling at a set up cost of Rs 1,00,000 and Rs 5 per unit of variable cost. A second design with spring action will have a fixed cost of Rs 1,60,000 and variable cost of Rs 4 per unit. Yet another design with weights and pulleys will have a fixed cost of Rs 3,00,000 and variable cost of Rs 3 per unit. One of the following demand levels and its probabilities can occur for the doll.

Demand Level	Demand (units)	Probability
Light demand	25,000	0.10
Moderate demand	1,00,000	0.70
Heavy demand	1,50,000	1.20

(i) Construct a pay off table for the above project.
(ii) Which is the optimum design?
(iii) How much can the decision maker afford to pay to obtain the perfect information about the demand? **(5M)**

(b) The mean nicotine content of a brand of cigarette is 20.0 mgs. A new process is proposed to lower the nicotine content without affecting the flavour. To test the new process, 16 cigarettes are selected at random from the week's output from the test plant. The sample mean nicotine content is found to be 18.5 mg. If the standard deviation of the nicotine contents is calculated to be 2mgs, is the claim of the new process justified? Use 5% level of significance. **(5M)**

Q5. (a) The following table gives the aptitude test scores and productivity indices of 10 workers selected at random:

Aptitude Scores(x)	60	62	65	70	72	48	53	73	65	82
Productivity Index(y)	68	60	62	80	85	40	52	62	60	81

Calculate the two regression equations and estimate
(i) The Productivity Index of a worker whose test score is 92.
(ii) The test score of a worker whose Productivity Index is 75. **(5M)**

(b) Most individuals are aware of the fact that the average annual repair cost for an automobile depends on the age of the automobile. A researcher is interested in finding out whether the variance of the annual repair cost also increases with the age of the automobile. A sample of 25 automobiles that are 4 years old showed a sample variance for annual repair cost of Rs 450 and a sample of 25 automobiles that are 2 years old showed a sample variance for the annual repair costs of Rs 300. Test the hypothesis that the variance in annual repair cost is more for the older automobile for a 0.01 level of significance.

(5M)

Q6. Read the following case and use the appropriate statistical tool to analyse the case. **(20M)**

THE CASE OF A MEDICAL REPRESENTATIVE

Mr. Makrand Godbole, sales representative of WKPIL (Well known Pharmaceuticals India limited) is one of the promising representatives located in Mumbai City. He has won several awards for his excellent job of meeting the targets. Last year, he has also won the National award of BEST REPRESENTATIVE of the year. One of the most important jobs of the medical representatives is to meet the practicing doctors and introduce to them some of their new products, and discuss with them about their advantages over the other existing products. As the awareness of the doctors about the products of WKPIL has a direct relation to the sales, the company fixes targets on the number of doctors to be visited over a period of time. WKPIL has a policy of finalizing the annual as well as quarterly targets in consultation with the concerned officials. The company believes that this is the best way of involving the entire organization in the decision making process, and it is observed that the officials become more accountable and are generally bound by the decision as they were part of the decision making process. To meet the current target, considering the number of visits that can be made per day, Mr. Godbole needs to meet 100 more doctors in Mumbai in the 27 days that is remaining in the quarter. The regional manager of Southern Region, Mr. Nagraj, has extended an invitation to Mr. Godbole to address and interact with his fellow representatives, in the current term, highlighting the factors that helped in his achievements. The company feels that this will be a motivating factor for other representatives. The venue for this meeting is identified as Bangalore, which is quite a distance from Mumbai. Even if he flies to Bangalore, Mr. Godbole needs a day exclusively for this purpose. Mr. Godbole knows that he requires at least 25 days to complete his target. As such, it looks it is possible to take a day off required to go to Bangalore. However, he is also aware that he cannot walk on a tight rope like this, because there are some of the days during which he cannot travel to meet the doctors due to the following exhaustive reasons.

a) In Mumbai City, a disturbing trend lately is that some political or social organizations announce bandh or hartal, as a mark of protest against some policy of the Government or to highlight a specific problem facing the society. During these days, there is a total restriction on movement of the public. And, therefore, during the days when a bandh or hartal is declared, Mr. Godbole will not be able to meet the doctors.

b) And also during the current season (viz. monsoon season) when it rains quite heavily some parts of the city gets flooded with water. As a result of this, some of the roads get blocked, suburban services are cancelled and, hence, on these days again Mr. Godbole will not be able to meet the doctors.

Since Mr. Godbole is not willing to miss the target, he wants to make sure that he works for at least 25 days to meet the target. At the same time, he is very keen to go to Bangalore to address his fellow workers in southern region, as this will be a professional boost to his career, and in the process, he may help his fellow workers also to excel. In order to ensure that he gets enough working days, he wishes to find out the frequency of the happenings of these two events. After scanning through the newspapers of the last two years, Mr. Godbole observed that during the monsoon there is a 1 in 30 chance that, on any day in this season, the roads are blocked due to flood in the city. He also observed from the records of the civic administration that the movement in the city was restricted due to bandh or hartal, etc. for 14 days in the last 2 years viz., about 730 days. What conclusion did Mr. Godbole arrive at? What are the methods Mr. Godbole used to arrive at this conclusion?

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